Jeremy Goldwasser

Website · Email · Linkedin · Google Scholar

Education

UC Berkeley, Berkeley, CA — Aug 2021 - May 2026 (Expected)
Ph.D. in Statistics; Advised by Ryan Tibshirani and Giles Hooker — GPA: 3.97

Yale University, New Haven, CT — Aug 2017 - May 2021
B.S. in Statistics and Data Science — GPA: 3.73

Interests

Interpretability; statistical ML; uncertainty quantification; time series; LLMs; AI for health.

Internship Experience

Apple — May - Aug 2024
Research Intern, Vision Products Group — Sunnyvale, CA

- Developed novel generative AI method for counterfactual image explanations.
- Tool is currently used to provide previously unattainable insights on failure modes of FaceID.

Genentech — May - July 2023
Data and Statistical Sciences Intern — South San Francisco, CA

- Produced confidence intervals for AI pathology models using conformal prediction.
- Calibrated cell-type classifier, quantifying uncertainty and improving test accuracy by 3%.

Voca.ai (acquired by Snapchat) — June - Aug 2019
Machine Learning Intern — Herzliya, Israel

- Trained Transformers for Automatic Speech Recognition and Named Entity Recognition.

Publications

1. **Jeremy Goldwasser** and Giles Hooker. "Unifying Image Counterfactuals and Feature Attributions with Latent-Space Adversarial Attacks." *International Conference on Machine Learning (ICML), Actionable Interpretability Workshop*, 2025.
2. **Jeremy Goldwasser** and Giles Hooker. "Statistical Significance of Feature Importance Rankings." *Conference on Uncertainty in AI (UAI)*, 2025.
3. **Jeremy Goldwasser**, Will Fithian, and Giles Hooker. "Gaussian Rank Verification." *Stat*, 2025.
4. **Jeremy Goldwasser** and Giles Hooker. "Stabilizing Estimates of Shapley Values with Control Variates." *World Conference on Explainability in AI (xAI)*, 2024.
5. Rui Yang, Qingcheng Zeng, Keen You, Yujie Qiao, Lucas Huang, Chia-Chun Hsieh, Benjamin Rosand, **Jeremy Goldwasser**, et al. "Ascle: A Python Natural Language Processing Toolkit for Medical Text Generation." *Journal of Medical Internet Research*, 2024.
6. Zhanlin Chen, **Jeremy Goldwasser**, Philip Tuckman, Jason Liu, Jing Zhang, and Mark Gerstein. "Forest Fire Clustering for Single-Cell Sequencing Combines Iterative Label Propagation with Parallelized Monte Carlo Simulations." *Nature Communications*, 2022.
7. Irene Li, Jessica Pan, **Jeremy Goldwasser**, et al. "Neural Natural Language Processing for Unstructured Data in Electronic Health Records: a Review." *Computer Science Review*, 2022.

Submissions

1. **Jeremy Goldwasser**, Addison J. Hu, Alyssa Bilinski, Daniel J. McDonald, and Ryan J. Tibshirani. "Estimating Time-Varying Epidemic Severity Rates with Adaptive Deconvolution." Preprint, October 2025. Under review at *Annals of Applied Statistics*.
2. **Jeremy Goldwasser**, Addison J. Hu, Alyssa Bilinski, Daniel J. McDonald, and Ryan J. Tibshirani. "Challenges in Estimating Time-Varying Epidemic Severity Rates from Aggregate Data." Preprint, December 2024. Under review at *PLOS Computational Biology*.

Ongoing Projects

- Attention-based interpretability of Vision Transformers for AI cancer pathology.
- Online learning for ensembling time series forecasters with uncertainty quantification.

Skills

Coursework: Deep Learning, Optimization, Causal Inference, AI in Biology
Programming: Python (PyTorch, NumPy, scikit-learn, Pandas, Matplotlib), R, LaTeX
Languages: Spanish (fluent), Hebrew (basic)

Misc Projects

AI Meets DNA Methylation May 2024

- Used interpretable ML to identify methylation sites that regulate CD55 gene expression.

Predicting Peptide-MHC Binding Affinity in SARS-CoV-2 April 2020

- Designed neural architecture to predict which viral peptides bind with immune cells.

Awards

U.S. Civic Digital Fellowship (Declined) 2021
Yale College Dean's Research Fellowship, Morse Richter Fellowship 2020
Yale Creative and Performing Arts Grant 2020

Invited Talks

- **"Stable Rankings for Explainable AI."** Berkeley Statistics Annual Research Symposium, September 2025.
- **"Estimating Epidemic Severity Rates."** UCSF MINDSCAPE, February 2025.
- **"Estimating Epidemic Severity Rates."** Delphi Group, January 2025.
- **"Estimation of Time-Varying Severity Rates from Aggregate Data: Pitfalls and New Ideas."** California Department of Public Health, September 2024.
- **"Estimation of Time-Varying Severity Rates from Aggregate Data: Pitfalls and New Ideas."** Centers for Disease Control and Prevention (CDC), September 2024.
- **"Stabilizing Estimates of Shapley Values with Control Variates."** 2nd World Conference on Explainable AI, Malta, July 2024.
- **"Provably Stable Feature Rankings with SHAP and LIME."** Apple Data Analytics and Quality Group, June 2024.
- **"Stable Feature Rankings & Time Series Deconvolution."** Genentech gRED Computational Sciences, February 2024.
- **"Provably Stable Feature Rankings with SHAP and LIME."** Center for Human-Compatible AI, January 2024.
- **"Provably Stable Feature Rankings with SHAP and LIME."** Berkeley Statistics Student Seminar, January 2024.
- **"Estimating Case Fatality Rate via Convolutional Modeling."** Delphi Group, April 2023.